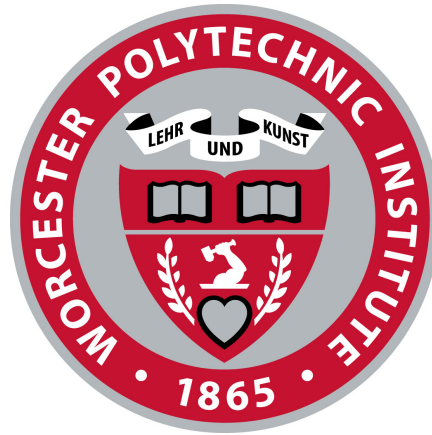




DS 2010

Modeling and Data

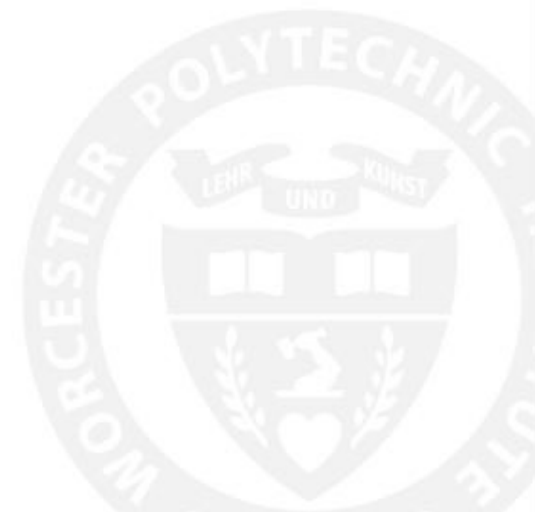
Analysis

Prof. Reza Zekavat (PhD)

Instructor Information

WPI

- **Office:** OH130
- **Email:** rezaz@wpi.edu
- Using email to solve problems is not suggested.
- Please use office hours for those questions.
- **Office hours:** Mondays **2:00 pm-3:00pm** or by appointment



Course Assistant Information

WPI

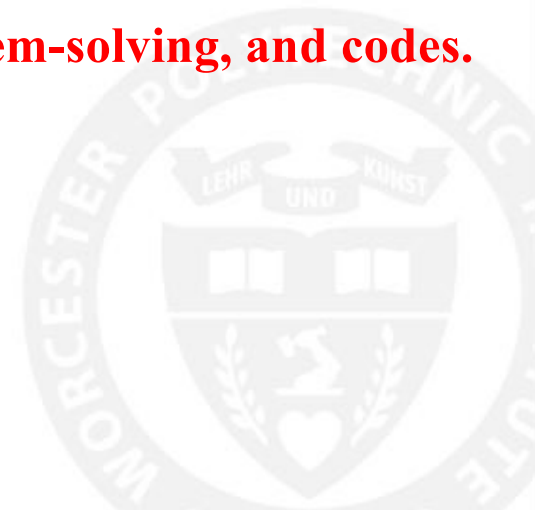
- **Graduate Learning Assistant (GLA): Vedant Kishor More**

Email: vmore@wpi.edu

Office Hours: Thursdays 3 – 5 pm

Location: TBD

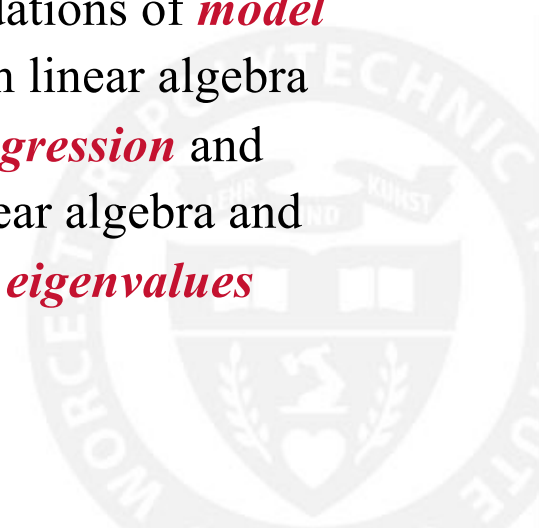
Note: The GLA will address your questions on grading, problem-solving, and codes.



Course Information



- **Course Number:** DS 2010
- **Course Name:** Modeling and Data Analysis
- **Meeting:** Mondays and Thursdays 12:00 - 1:50 pm
- **Location:** UH420
- **Course Description:** This course focuses on the mathematical foundations of *model and data-driven approaches* in Data Science. It covers methods from linear algebra and applied statistics that are key to data analysis methods such as *regression* and *machine learning*. Topics introduced in this course include basic linear algebra and statistics, basics of machine learning (*classification and clustering*), *eigenvalues* and *singular matrices*, data exploration, and data cleaning.



The Goal of this Course



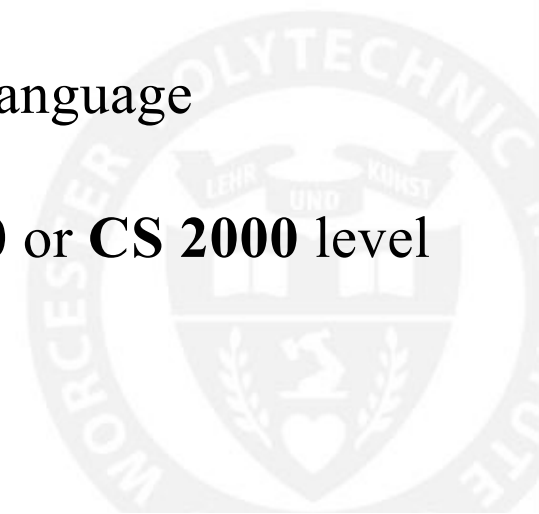
To learn mathematical and statistical models that support us to:

- Draw inferences from data
- Make data-driven predictions
- Optimize decision-making using mathematical frameworks
- Understand and quantify uncertainty in data
- Identify patterns and trends through statistical analysis
- Apply probability theory to model real-world scenarios, perform hypothesis testing to validate insights

Recommended Background



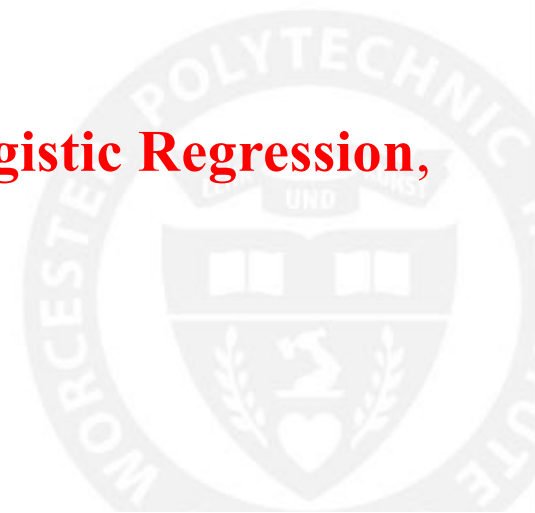
- Data science basics equivalent to **DS 1010**
- Applied statistics and regression equivalent to **MA 2611** and **MA 2612**
- The ability to write computer programs in a scientific language
equivalent to a CS programming course at the **CS 1000** or **CS 2000** level



Learning Objectives

WPI

- Understanding key concepts in **linear algebra** and applying them in data science.
- Understanding key concepts in **probability and statistics**.
- Using tools such as **Classification, Linear Regression, Logistic Regression**, etc. for predictions.
- Explaining the pros and cons of various approaches.



Computing Background for the Course



- You will need some background playing with, processing, and plotting data with the Python programming language
 - Some homework problems use Python
 - One of our two textbooks uses Python
 - Course TA will support you with your Python needs
- Note: this is not a programming course (i.e., your code will not be graded), but working with data will be very important (i.e., the results of the code will be graded).

Python



-
- Python can be run “in the cloud” on Google Colab
 - You can set up a Python account for free at <https://research.google.com>
 - Or you can download Python3 at <https://www.anaconda.com/>
 - You will also likely want to install Jupyter Notebook, a popular interface for Python
 - Jupyter Notebook can be downloaded at <https://jupyter.org/install>
 - Detailed instructions on installing Python (or running Python on Google Colab) are given on page 40 of our textbook

Your feedback is highly valued!

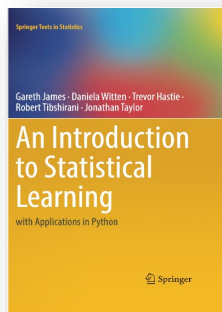
- Your feedback is **highly valued**, and
- It will help improve the course :-)

Textbook

WPI

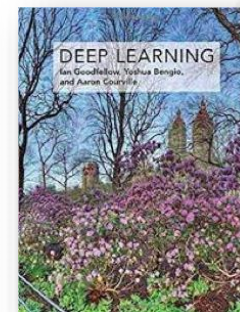
An Introduction to Statistical Learning With Applications in Python

- **Authors:** Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor
- Available Online at: <https://www.statlearning.com/>



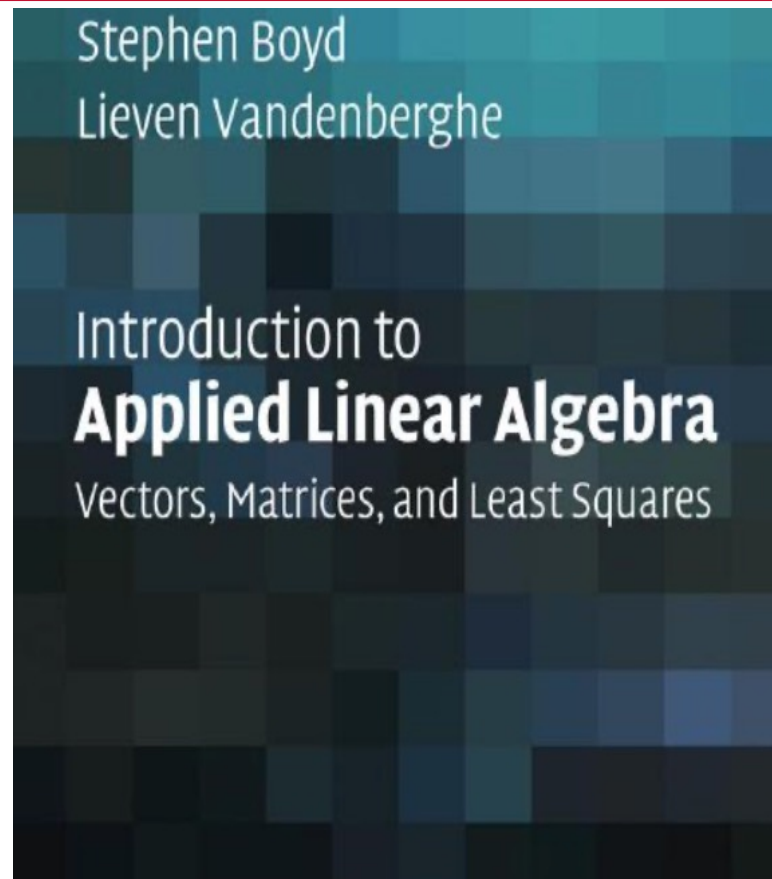
Deep Learning

- **Authors:** Ian Goodfellow, Yoshua Bengio, Aaron Courville
- Available on authors' website, or you can also purchase a physical copy.



More Resources

WPI



More Resources

WPI

O'Reilly Books on Data Science

- **Available on:**

<http://go.oreilly.com/worcester-polytechnic-institute>

- You can use your WPI credentials to access many O'Reilly books on data science.

O'REILLY®

Python Resources

Official Python3 Documentation

- **Available on:**

[3.9.6 Documentation \(python.org\)](https://docs.python.org/3.9.6/)

- Available on authors' website, or you can also purchase a physical copy.



More Resources



Other textbooks that would be useful for the course are:

- Linear Algebra and Its Applications, by Gilbert Strang
- Applied Statistics for Engineers and Scientists, by Joseph Petrucci, Balgobin Nandram, and Minghui Chen
- This has been the textbook for MA2611 and MA2612 (one of the requirements for the course)
- Good books on Python for Data Science and Machine Learning:
 - **“Python for Data Analysis”**, Wes McKinney
 - **“Python Data Science Handbook”**, Jake VanderPlas
 - **“Python Data Science Essentials”**, Alberto Boschetti and Luca Massaron

Grading Rubric



Usually:

A: > 90%

B: 80% – 90%

C: 65% - 80%

NR: < 65%

Components	Percentage
Assignments	25%
Midterm Exam	25%
Final Project	25%
Final Exam	25%
Quiz (Extra Points)	10%
Total	110%

Assignment Schedule and Policy



Assignments	Due Friday
Assignment 1	Jan 30
Assignment 2	Feb 06
Assignment 3	Feb 20
Assignment 4	Feb 27

Late Assignment policy

- In general, late assignments will either not be accepted, or:
 - Late assignments within 24 hours are subject to 50% deduction
 - Late assignments after 24 hours will be graded but subject to 100% deduction
- If an emergency arises or you know in advance about a conflict, please let me know as soon as possible.

Exam Schedule and Policy



Exams	Date
Midterm exam	Feb 12
Final exam	Mar 02

Exam Notes:

1. The midterm exam and final exam will be during class time.
2. Exams will be proctored by the course TA and Grader.
3. Students can bring up to two (2) 8 ½ by 11 sheets of paper (either printed or handwritten) with whatever notes you want for the exam.

Important Exam Logistics

1. You need to arrive 5 min before the exam time. The exam will start at 12:00 pm sharp.
2. You should maintain a proper space between you and the neighboring student.
3. We may adjust students' seats in the middle of the exam if students are too close.
4. You can bring your Calculator and 2-page Cheat Sheet to the Exam.
5. You are not allowed to use any other resources.
6. Turn your WiFi off on all devices.
7. Computers are not allowed in the exam.

Project Schedule and Policy



Project	Due
Abstract Draft	Feb 09
Abstract Presentation	Feb 16
Final Presentations	Mar 05
Final Reports	Mar 06

Notes:

- You will also be allowed to select your own teams of 2-5 for the final project.
- The number of team members should be consistent with the project size and complexity.
- All team members should represent a contribution.
- On the final project, each team of students will use their own data sets.
- The project proposal (due April 15) includes the project title, team members, and abstract.
- Abstract should highlight the data, planned analysis, and the value of your project.

Project Grading



Project	Percentage
Proposal	5%
Report	10%
Presentation	10%

Notes:

- Each team will present their proposal only for 2 min in the classroom.
- The proposal should highlight the value of the project and the contribution of each member.
- Presentations should highlight: (a) Motivation: application and why data is important, (b) data analysis stages, (C) results, and (D) conclusions.
- Each team will present the final report in the class.
- Presentations will be rated by the course TA and Grader.

Project Abstract Elements



1. The project title.
2. Team members.
3. Project Goal.
4. Who will do what to cover the project goal?
5. Timeline.

Project Presentation Content



1. Motivation: Why the problem associated to Data is important (1 Slide).
2. What Data has been used? (1 Slide).
3. What Data Analysis Methods have you used and Why you adopted them? (1 - 2 Slides).
4. What performance measures have you used? (1 Slide).
5. Results and a comparison across Data Analysis methods used. (2 - 5 Slides).
6. Concluding Remarks. (1 Slide).

Tips on Good Presentation

1. Organization in slides, in mind and in presentation;
2. Exercise well to make sure you: (a) don't say any additional sentence which is not required, and (b) accommodate all info within the assigned time frame;
3. Don't stay in a place without any movement. Have a movement in harmony with your presentation;
4. Do not put your hand in your pocket;
5. Speak loud enough and make connection with eyes of audience;
6. Use as short sentences as possible in your slides;
7. Avoid writing long equations as much as possible. Present all parameters of an equation in your slide clearly. So, the audience can see and read each parameter;
8. Use as many figures as possible to clear your presentation;
9. Always assume that the audience does not know all principles of the work. Hence, you need to assign enough number of slides to present the principles of the work before going through the details of the work;
10. Clear up all the curves in your results by presenting each curve with an arrow;
11. In the simulation results, explain the axes before introducing the curves;
12. Present the main conclusion of each simulation in the same slide you present the simulation results;
13. Do not go with abbreviations as much as possible, and present the full word unless you are sure it is clear to everybody;
14. Don't use font sizes less than 18;

Collaboration and Academic Honesty Policy **WPI**

- Collaboration is encouraged on assignments and the final project.
- On assignments you **may** discuss problems with other students, but each student will be responsible for generating solutions and writing up results **from scratch**.

Collaboration



I want to emphasize the importance of collaboration and accountability **in team projects**. When forming a team, it's essential that all members are assigned **clear responsibilities**. However, please be aware that if any team member fails to fulfill their tasks, **the entire team's grade will be impacted** — we do not assign individual grades within group projects. Therefore, **choose your teammates wisely**.

To ensure fair contribution and harmony within the team, all members will be asked to complete a peer review survey at the end of the project. For teams with more than two members, it is highly recommended to designate a team leader who will coordinate workload distribution and set timelines for completing tasks.

Regarding assignments, the expectations differ slightly. As outlined in the syllabus and discussed in class, all team members must collaboratively work on every problem. Dividing the assignment by assigning different problems to different teammates is not allowed. Regardless of internal task sharing, the assignment will be graded as a single submission, and the grade will apply to the entire team — **no individual grading will be done**.

Collaboration and Academic Honesty Policy



-
- *As examples*, each of the following would **be a violation of** the collaboration policy (this list is **not** exhaustive):
 - Two different students share an assignment's solution to any assigned problem.
 - One assignment or project team allows another assignment or project team to copy any part of a solution to an assigned problem.
 - Any code or plots are shared between assignments or project teams.
 - *As examples*, each of the following would **not be a violation** of the collaboration policy:
 - Students within a team sharing solutions and code for a problem.
 - Students from different teams discussing an assignment at the level of goals, where ideas for solutions can be found in the book or notes, what parts are more challenging, or how one might approach the problem.
 - You can ask me or the TA any questions you like.
 - If there is any doubt as to what is allowed and what is not allowed, please just ask!

Student Responsibilities and Course Policies



- **Accommodation for Special Needs or Disabilities**

If you need course adaptations or accommodations because of a disability, or if you have medical information to share with me, please make an appointment with me as soon as possible.

If you have not already done so, students with disabilities who believe that they may need accommodations in this class are encouraged to contact the Office of Accessibility Services as soon as possible to ensure that such accommodations are implemented in a timely fashion.

- **Accommodation for Religious Observance**

Students requiring accommodation for religious observance must make alternate arrangements with Prof. Zekavat at least one week before the date in question.

Student Responsibilities and Course Policies



- Personal Emergencies

- In the event of a medical or family emergency, please contact me to work out appropriate accommodations.
- Other than personal emergencies, please let me know at least 1 week in advance

- Make-up exam policy

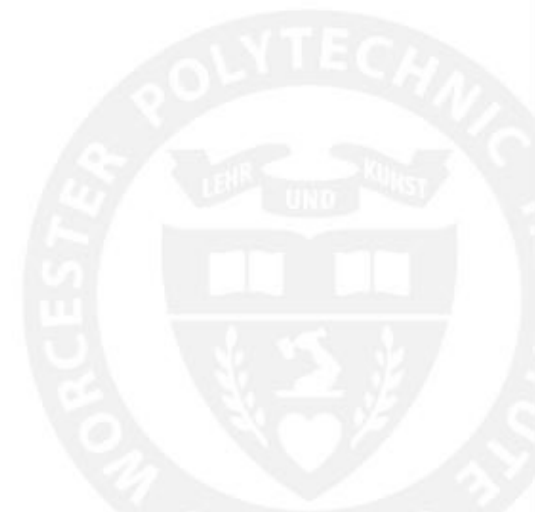
- Make-up exams will only be allowed in the event of a documented emergency or religious observance.
- The exam dates are listed on the syllabus, and you are responsible for avoiding conflicts with the exams.

Attendance and Participation

WPI

You are expected to:

- Attend class
- Arrive on time
- Turn off/silence your cell phones
- Participate in class
- Enjoy learning



Success Keys



The most important part of the learning experience:

- Solving the assigned homework
- Attending the lectures
- Asking questions. Questions are an important part of the learning process.
- Any question you ask will help your classmates as much as you.
- The slides will be posted on: canvas.wpi.edu

I wish you a healthy, happy, productive, and successful term.

